

Student–Teacher–Subject Code

AIM

To write and execute a Prolog program to represent the relationship between students, teachers and subjects using a knowledge base.

ALGORITHM

1. Start the program.
2. Define facts representing the subjects handled by teachers.
3. Define facts representing the subjects studied by students.
4. Use Prolog rules to determine the teacher corresponding to a student's subject.
5. Query the database to obtain the required student, subject and teacher details.
6. Display the result.
7. Stop the program.

PROGRAM

% Student-Teacher-Subject database

teacher(john, mathematics).

teacher(mary, physics).

teacher(david, chemistry).

teacher(steve, computer_science).

student(rahul, mathematics).

student(anu, physics).

student(kiran, chemistry).

student(riya, computer_science).

student_teacher(Student, Teacher) :-

student(Student, Subject),

teacher(Teacher, Subject).

OUTPUT

```
11 ?- consult(['C:/Users/Lokesh/Desktop/17-33/student_teacher.pl']).
true.

12 ?- student_teacher(rahul,Teacher).
Teacher = john.

13 ?- student_teacher(Student,Teacher).
Student = rahul,
Teacher = john ▲
```

RESULT

Thus, the Prolog program to represent and retrieve Student–Teacher–Subject relationships was successfully executed, and the required relationships were obtained successfully.